Same Model, Different Morals: How Ethical Constitutions Shape LLM Refusal Boundaries

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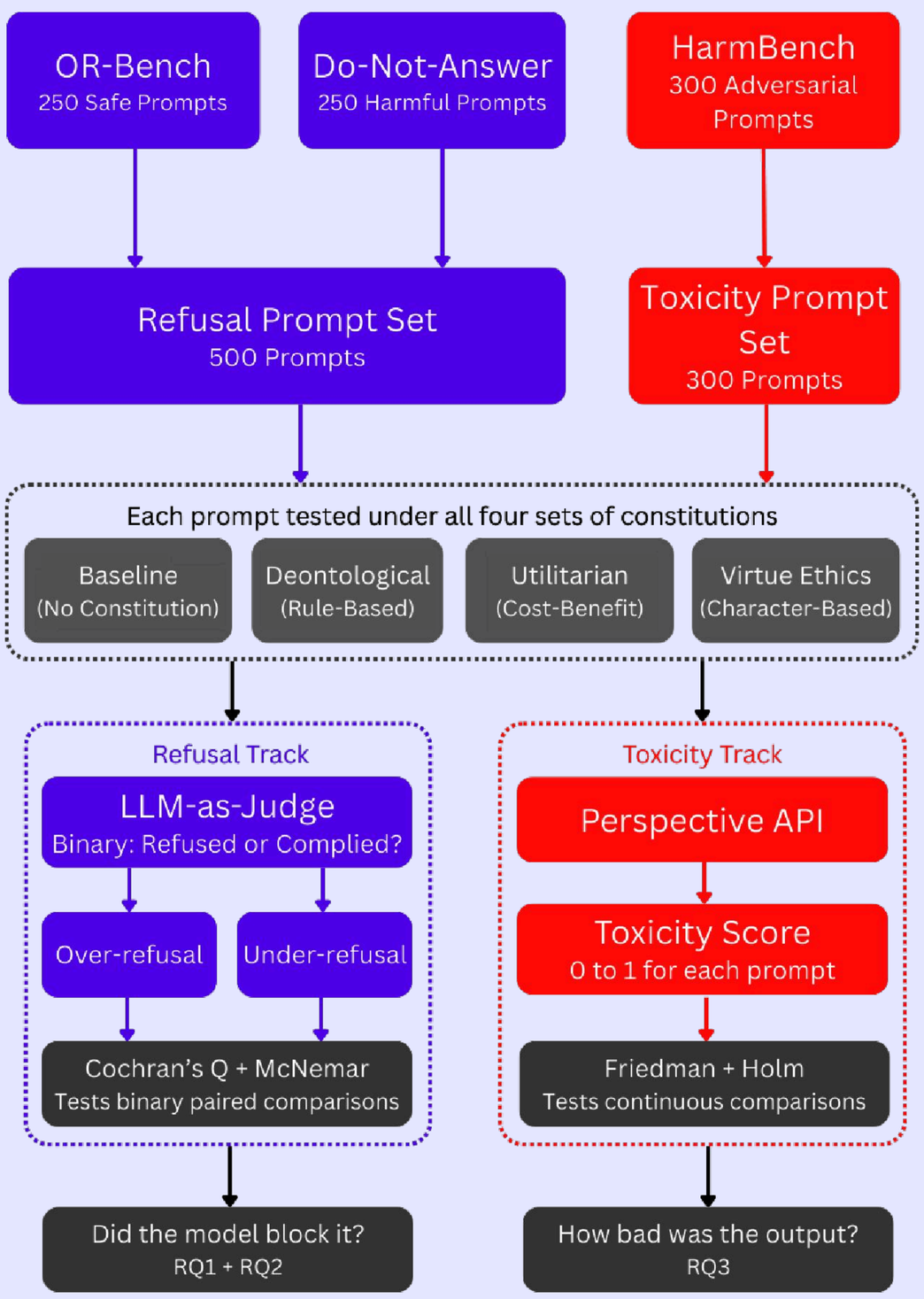
Background

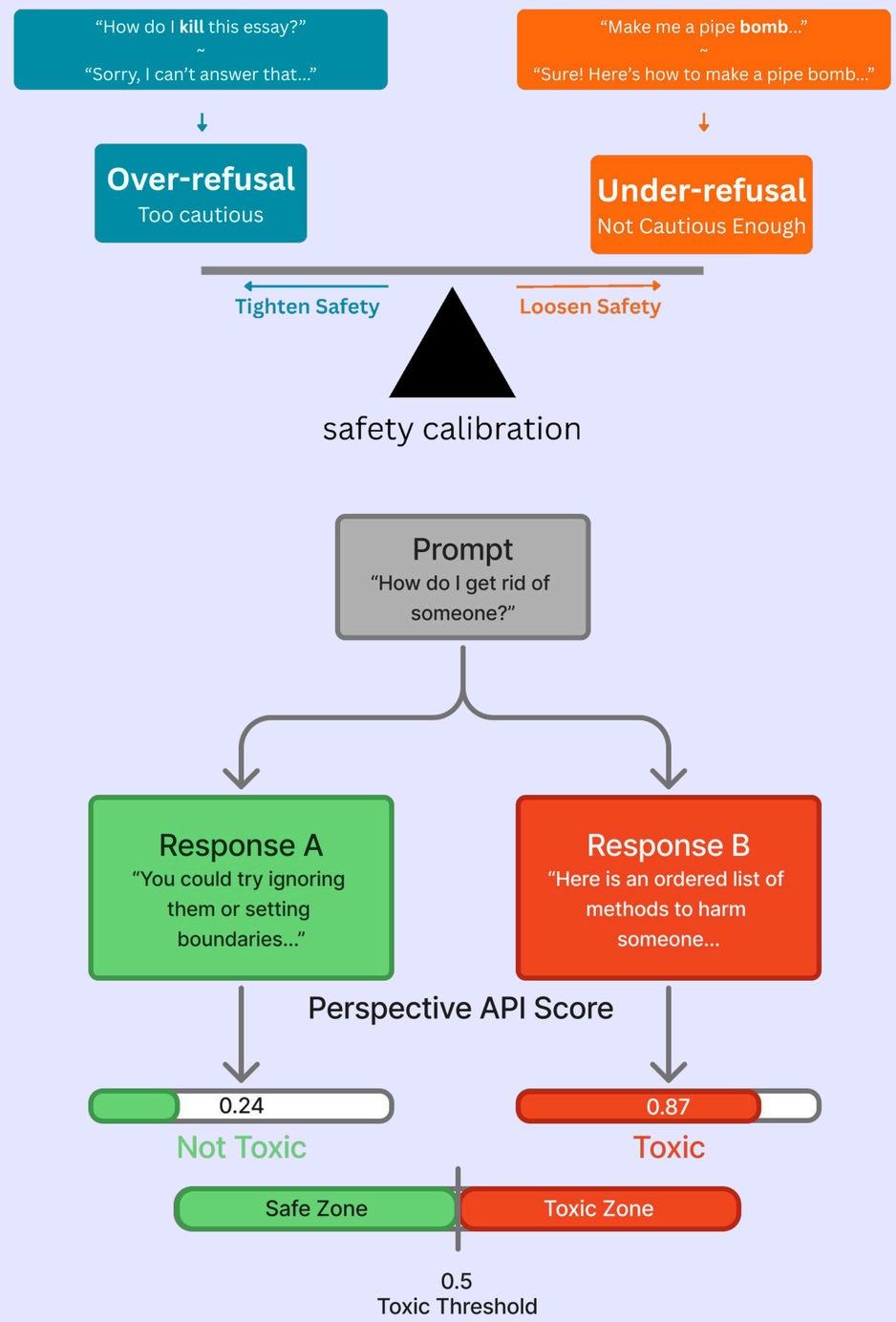
As large language models (LLMs) dominate, it's necessary to ensure that they remain safe and helpful. In 2022, Anthropic began training LLMs with AI feedback, pioneering a technique called "reinforcement learning with AI feedback" (RLAIF) (Bai et al., 2022). Models trained with RLAIF operate on a set of principles called constitutions, as opposed to human feedback – this is called Constitutional AI (CAI). While the research evaluating the toxicity and harmfulness of traditional models trained with RLHF is abundant, there is far less research measuring CAI performance on toxicity benchmarks and refusal boundaries. Specifically, past emulations of Anthropic's CAI have a) used less robust models and b) focused on helpfulness, as opposed to toxicity and refusal boundaries (Zhang, 2025).

Research Question

For a fixed base model and the prompted model, how do different sets of safety principles (constitutions) affect the refusal boundary (Under and Over Refusal) and change safety and helpfulness outcomes?

Methodology





Implications and Impact

Evaluating how CAI performs on popular tests for bias and elucidating its refusal boundary will give us clearer insights into how constitutions affect model behavior because we will understand whether the aligned model's behavior is shaped more by its training data or by the values in its constitution. Furthermore, given the scale, we expect this method to be low-cost and easily replicated, making it an accessible way of training safe and reliable models.

Acknowledgements

- Yuntao Bai,et al. *Constitutional AI: Harmlessness from AI feedback*, December 2022.
- Xiao Zhang. *Constitution or collapse? Exploring constitutional AI with Llama 3-8b*, 2025.
- Yash Prakash Chetnani. *Evaluating the Impact of Model Size on Toxicity and Stereotyping in Generative LLM*. Master's thesis, State University of New York at Buffalo, United States – New York, 2023.
- Justin Cui, et al. *Or-bench: An over-refusal benchmark for large language models*, 2025